

Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
 MATH 455: Partial Differential Equations

Assignment #3 - Wave Equation
 Due Tuesday, October 6, 2026

1. (15 points) Define the functions

$$f(x) = \begin{cases} x, & \text{if } 0 \leq x \leq 1 \\ 1, & \text{if } 1 < x < 5 \\ 6 - x, & \text{if } 5 \leq x \leq 6 \\ 0, & \text{else,} \end{cases}$$

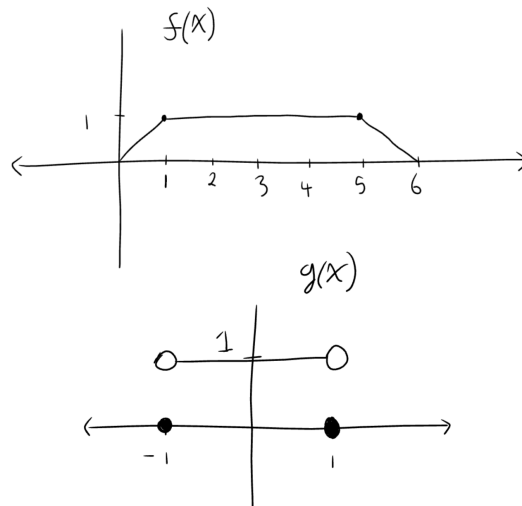
$$g(x) = \begin{cases} 1, & \text{if } |x| < 1 \\ 0, & \text{else,} \end{cases}$$

and consider solutions of the wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0. \quad (\text{WE})$$

- (a) Graph $f(x)$ and $g(x)$ by hand.

Solution —



(b) Find the solution of (WE) for $x \in \mathbb{R}, t > 0$ with

$$u(0, x) = f(x) \quad \text{and} \quad \frac{\partial u}{\partial t}(0, x) = 0$$

for $x \in \mathbb{R}$. In particular, write the solution in terms of the $f(x)$ function. Then, graph by hand $u(\frac{1}{c}, x)$, $u(\frac{2}{c}, x)$, and $u(\frac{3}{c}, x)$ using different axes.

Solution — To find the general solution of (WE), apply the substitution

$$\begin{aligned} r(t, x) &= x - ct \\ s(t, x) &= x + ct \end{aligned}$$

Assume $u(t, x) = U(r, s)$ is a solution and substitute (we first calculate the needed derivatives of u):

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial U}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial U}{\partial s} \frac{\partial s}{\partial x} \\ &= \frac{\partial U}{\partial r} + \frac{\partial U}{\partial s} \end{aligned}$$

And

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial U}{\partial r} + \frac{\partial U}{\partial s} \right)$$

To simplify further, note that $\frac{\partial U}{\partial r}$ is some function of r and s , say $\phi_1(r, s)$, and $\frac{\partial U}{\partial s}$ is also some function of r and s , say $\phi_2(r, s)$:

$$\begin{aligned} &= \frac{\partial}{\partial x} (\phi_1(r, s) + \phi_2(r, s)) \\ &= \frac{\partial \phi_1}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial \phi_1}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial \phi_2}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial \phi_2}{\partial s} \frac{\partial s}{\partial x} \\ &= \frac{\partial \phi_1}{\partial r} + \frac{\partial \phi_1}{\partial s} + \frac{\partial \phi_2}{\partial r} + \frac{\partial \phi_2}{\partial s} \\ &= \frac{\partial}{\partial r} \left(\frac{\partial U}{\partial r} \right) + \frac{\partial}{\partial s} \left(\frac{\partial U}{\partial r} \right) + \frac{\partial}{\partial r} \left(\frac{\partial U}{\partial s} \right) + \frac{\partial}{\partial s} \left(\frac{\partial U}{\partial s} \right) \\ &= \frac{\partial^2 U}{\partial r^2} + \frac{\partial^2 U}{\partial s \partial r} + \frac{\partial^2 U}{\partial r \partial s} + \frac{\partial^2 U}{\partial s^2} \end{aligned}$$

Let us assume that $U(r, s)$ is smooth and well-behaved enough for the cross-derivatives to be equal. Then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 U}{\partial r^2} + 2 \left(\frac{\partial^2 U}{\partial r \partial s} \right) + \frac{\partial^2 U}{\partial s^2}$$

By similar mechanics, we obtain

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 U}{\partial r^2} - 2 \left(\frac{\partial^2 U}{\partial r \partial s} \right) + \frac{\partial^2 U}{\partial s^2} \right)$$

Substituting these second derivatives in the general wave equation yields:

$$\begin{aligned} & \left[c^2 \left(\frac{\partial^2 U}{\partial r^2} - 2 \left(\frac{\partial^2 U}{\partial r \partial s} \right) + \frac{\partial^2 U}{\partial s^2} \right) \right] \\ & - c^2 \left[\frac{\partial^2 U}{\partial r^2} + 2 \left(\frac{\partial^2 U}{\partial r \partial s} \right) + \frac{\partial^2 U}{\partial s^2} \right] \\ & = -4c^2 \left(\frac{\partial^2 U}{\partial r \partial s} \right) = 0 \\ & \iff \frac{\partial^2 U}{\partial r \partial s} = 0 \end{aligned}$$

Our change-of-variables has transformed the original wave equation into this more workable form, which we can now simply integrate to extract U :

$$\frac{\partial}{\partial r} \frac{\partial U}{\partial s} = 0$$

$$\iff \frac{\partial U}{\partial s} = C_1(s)$$

$$\iff U = C_2(s) + C_3(r)$$

$$\iff U(r, s) = F(r) + G(s)$$

$$\iff u(t, x) = F(x - ct) + G(x + ct)$$

We now apply our initial conditions.

$$u(0, x) = f(x) \implies F(x) + G(x) = f(x)$$

$$\begin{aligned} \left. \frac{\partial u}{\partial t} \right|_{(0, x)} = 0 &\implies -cF'(x) + cG'(x) = 0 \\ &\implies F'(x) = G'(x) \\ &\implies F(x) = G(x) + \delta \quad \text{for some } \delta \in \mathbb{R} \end{aligned}$$

Our two constraints combine to imply

$$\begin{aligned} 2G(x) + \delta &= f(x) \\ \implies G(x) &= \frac{1}{2}(f(x) - \delta) \\ \implies F(x) &= \frac{1}{2}(f(x) + \delta) \end{aligned}$$

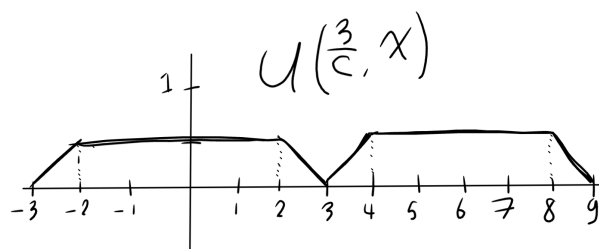
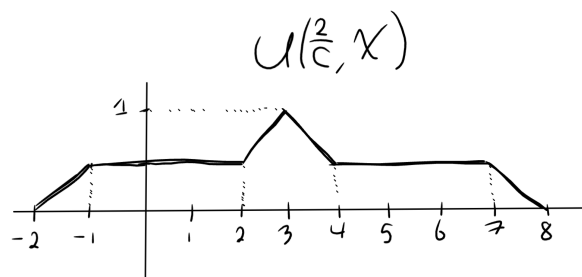
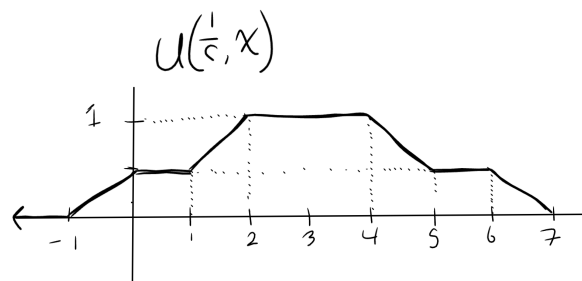
Hence

$$\begin{aligned} u(t, x) &= \frac{1}{2}(f(x - ct) + \delta) + \frac{1}{2}(f(x + ct) - \delta) \\ &= \boxed{\frac{1}{2}f(x - ct) + \frac{1}{2}f(x + ct)} \end{aligned}$$

$$\boxed{\begin{aligned} u\left(\frac{1}{c}, x\right) &= \frac{1}{2}f(x - 1) + \frac{1}{2}f(x + 1) \\ u\left(\frac{2}{c}, x\right) &= \frac{1}{2}f(x - 2) + \frac{1}{2}f(x + 2) \\ u\left(\frac{3}{c}, x\right) &= \frac{1}{2}f(x - 3) + \frac{1}{2}f(x + 3) \end{aligned}}$$

(c) Find the solution of (WE) for $x \in \mathbb{R}, t > 0$ with

$$u(0, x) = 0 \quad \text{and} \quad \frac{\partial u}{\partial t}(0, x) = g(x)$$



for $x \in \mathbb{R}$. In particular, write the solution in terms of the $g(x)$ function. Then, graph by hand $u(\frac{1}{c}, x)$, $u(\frac{2}{c}, x)$, and $u(\frac{3}{c}, x)$ using different axes.

Solution — Now we apply different initial conditions to the same general solution we obtained $u(t, x) = F(x - ct) + G(x + ct)$:

$$u(0, x) = 0 \implies F(x) = -G(x)$$

$$\left. \frac{\partial u}{\partial t} \right|_{(0, x)} = g(x) \implies -cF'(x) + cG'(x) = g(x)$$

From our first constraint, $F'(x) = -G'(x)$, so

$$-c(-G'(x)) + cG'(x) = g(x)$$

$$G'(x) = \frac{g(x)}{2c}$$

$$G(x) = \frac{1}{2c} \int g(x) dx + \delta$$

$$F(x) = -\frac{1}{2c} \int g(x) dx - \delta$$

Let $A(x)$ be the antiderivative of g , so that $A'(x) = g(x)$:

$$G(x) = \frac{1}{2c} A(x) + \delta \quad F(x) = -\frac{1}{2c} A(x) - \delta$$

Hence

$$\begin{aligned} u(t, x) &= \left(-\frac{1}{2c} A(x - ct) - \delta \right) + \left(\frac{1}{2c} A(x + ct) + \delta \right) \\ &= \frac{1}{2c} \left(A(x + ct) - A(x - ct) \right) \\ &= \boxed{\frac{1}{2c} \int_{x-ct}^{x+ct} g(\zeta) d\zeta} \end{aligned}$$

We compute a few solution functions at various times. First, note that the area under g from a to b is

$$A(x) = \begin{cases} 0, & \text{if } x < -1 \\ x + 1, & \text{if } |x| < 1 \\ 2, & \text{if } x > 1 \end{cases}$$

Now, at different time increments $\frac{n}{c}$ for $n = 1, 2, 3 \dots$, the solution is

$$\begin{aligned}
u\left(\frac{n}{c}, x\right) &= \frac{1}{2c} \int_{x-n}^{x+n} g(\zeta) d\zeta \\
&= \frac{1}{2c} \left(A(x+n) - A(x-n) \right) \\
&= \frac{1}{2c} \left(\begin{cases} 0, & \text{if } x+n < -1 \\ (x+n)+1, & \text{if } |x+n| < 1 \\ 2, & \text{if } x+n > 1 \end{cases} - \begin{cases} 0, & \text{if } x-n < -1 \\ (x-n)+1, & \text{if } |x-n| < 1 \\ 2, & \text{if } x-n > 1 \end{cases} \right) \\
&= \frac{1}{2c} \begin{cases} 0 & \text{if } x \leq -n-1 \\ n+1+x, & \text{if } -n-1 \leq x \leq -n+1 \\ 2, & \text{if } -n+1 \leq x < n-1 \\ n+1-x, & \text{if } n-1 \leq x \leq n+1 \\ 0, & \text{if } x \geq n+1 \end{cases}
\end{aligned}$$

Consider the symmetry of the $n+1+x$ and $n+1-x$ cases. If we apply the substitution $A = n-1$ and $B = n+1$, then these cases reduce to

$$\begin{cases} B+x & \text{if } -B \leq x \leq -A \\ B-x & \text{if } A \leq x \leq B \end{cases}$$

Which reduces to

$$\begin{cases} B-|x| & \text{if } A \leq |x| \leq B \end{cases}$$

(Note that $B+x = B-|x|$ when x is negative (in the first interval) and $B-x = B-|x|$ when x is positive (in the second interval)). Also note that the two zero cases can be described by $|x| \geq n+1$. Hence

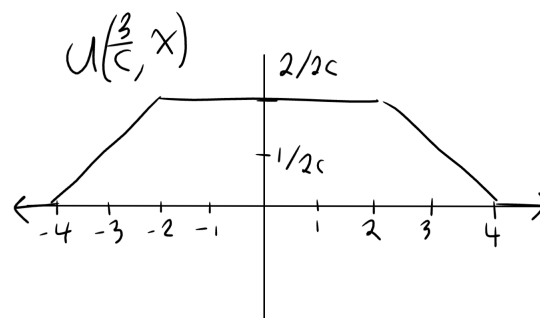
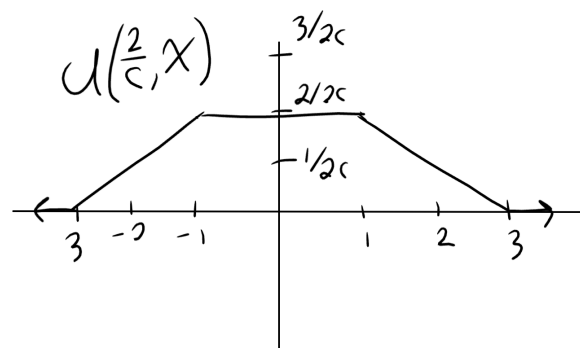
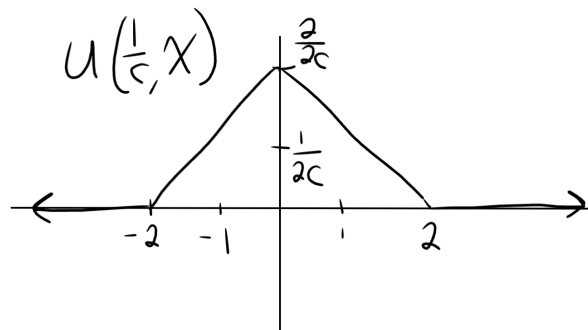
$$u\left(\frac{n}{c}, x\right) = \frac{1}{2c} \begin{cases} 0 & \text{if } |x| \geq n+1 \\ n+1-|x| & \text{if } n-1 \leq |x| \leq n+1 \\ 2 & \text{if } |x| \leq n-1 \end{cases}$$

Hence

$$u\left(\frac{1}{c}, x\right) = \frac{1}{2c} \begin{cases} 0 & \text{if } |x| \geq 2 \\ 2-|x| & \text{if } 0 \leq |x| \leq 2 \end{cases}$$

$$u\left(\frac{2}{c}, x\right) = \frac{1}{2c} \begin{cases} 0 & \text{if } |x| \geq 3 \\ 3-|x| & \text{if } 1 \leq |x| \leq 3 \\ 2 & \text{if } |x| \leq 1 \end{cases}$$

$$u\left(\frac{3}{c}, x\right) = \frac{1}{2c} \begin{cases} 0 & \text{if } |x| \geq 4 \\ 4 - |x| & \text{if } 2 \leq |x| \leq 4 \\ 2 & \text{if } |x| \leq 2 \end{cases}$$



2. (12 points) Let $f, g \in PC^1[0, L]$ be given continuous functions. Find the solution of (WE) for $0 < x < L$ and $t > 0$ with the boundary conditions

$$\frac{\partial u}{\partial x}(t, 0) = \frac{\partial u}{\partial x}(t, L) = 0, \quad t > 0$$

and the initial conditions for $0 < x < L$

$$u(0, x) = f(x), \quad \frac{\partial u}{\partial t}(0, x) = g(x).$$

Solution —

3. (15 points) Assume $u(t, x)$ satisfies (WE) for $0 < x < L$ and $t > 0$ and define

$$\mathcal{E}(t) = \frac{1}{2} \int_0^L \left| \frac{\partial u}{\partial t}(t, x) \right|^2 dx + \frac{1}{2} c^2 \int_0^L \left| \frac{\partial u}{\partial x}(t, x) \right|^2 dx$$

to be the associated total energy, where the first term in $\mathcal{E}(t)$ represents the kinetic energy and the second represents the potential energy.

- (a) Use **integration by parts** to show that the total energy satisfies

$$\mathcal{E}'(t) = c^2 \frac{\partial u}{\partial t}(t, x) \frac{\partial u}{\partial x}(t, x) \Big|_{x=0}^{x=L}.$$

Solution —

- (b) Show that the energy is conserved, meaning $\mathcal{E}(t) = \mathcal{E}(0)$ for all $t \geq 0$, for the zero Neumann boundary condition, namely assume that for $t \geq 0$

$$\frac{\partial u}{\partial x}(t, 0) = 0 \quad \text{and} \quad \frac{\partial u}{\partial x}(t, L) = 0.$$

Solution —

- (c) Show that the energy is conserved for any Dirichlet boundary conditions, namely for given $A, B \in \mathbb{R}$ assume that for $t \geq 0$

$$u(t, 0) = A \quad \text{and} \quad u(t, L) = B.$$

Solution —

- (d) Show that the potential energy and kinetic energy are equal for a traveling wave solution

$$u(t, x) = v(x - ct)$$

where v is a sufficiently smooth function.

Solution —

4. (8 points) Maxwell's equations for electric $E(t, x)$ and magnetic $B(t, x)$ fields in a vacuum are

$$\begin{cases} \frac{\partial B}{\partial t} + \nabla \times E = 0 \\ \mu_0 \epsilon_0 \frac{\partial E}{\partial t} - \nabla \times B = 0 \\ \nabla \cdot E = \nabla \cdot B = 0 \end{cases}$$

where $\mu_0 > 0$ is the permeability of free space and $\epsilon_0 > 0$ is the permittivity of free space. Using the “curl-curl” identity

$$\nabla \times (\nabla \times \phi) = \nabla (\nabla \cdot \phi) - \Delta \phi$$

for any $\phi(x)$, show that both E and B satisfy the (vector) wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \Delta u = 0$$

and determine the associated speed of propagation c . Here, $\Delta \phi = \nabla \cdot (\nabla \phi)$ is the Laplacian of ϕ , which represents a generalization of the second derivative in higher dimensions.

Solution —